
CSC3209 - Software Architecture and Design Pattern – Practical 2

- 1- The lecture argues that architecture brings a number of tangible benefits. How would you measure the benefits, on a particular project, of five of the thirteen points?

Sample Solution:

- **Inhibiting or enabling a system's quality attributes** – measure the benefits by achieving the system's quality attributes, in terms of performance, security and reusability. Performance can be measured by measuring time-based behaviour of the elements, frequency and volume of inter-element communication. Other than that, security can be measured by measuring inter-element communication. To measure reusability, we can measure the degree of coupling of elements. Good architecture enables low coupling and high cohesion. For example, in Sunway eLearn's architecture, the inter-element coupling between tools modules is low so that they can be highly reusable in various environment. For security, a proper authorization mechanism is applied in the eLearn system for certain user to access certain information so that the lecturers can set the permission for their subjects. Hence, the system is secured because information will not be easily exposed to unauthorized users.
- **Reasoning about and managing change** – measure amount of local and non-local changes. Good architecture benefits the design by having most changes to be local changes. The eLearn's architecture will be the basis for changes after deployment. Architecture shows the element, elements' interaction, and their architectural approach. Every design is reasoned in the architecture. When accommodating new features, architect will refer to the architecture to decide the possible changes. For example, in Sunway eLearn's architecture, local change is modifying a single element such as the live session feature without affecting other features, non-local change is modifying multiple elements such as database elements and authorization control elements, and architectural change such as the entire system structure and data model. It is better to have local changes rather than non-local and architectural changes.

- **Predicting system qualities** – architecture serves as a guide to predict system qualities as we reason the quality attributes in the architecture. This can help in making decisions which can be measured with the cost, time and amount of risk. For example, when designing eLearn’s database architecture model, system security could be predicted early in the software development life-cycle referring to the architecture. If we specify the system architecture considering different types of users (admin, lecturer, students) with different level of permissions, we can use this information to predict the system security in later phases such as the implementation level.
- **Enhancing communication among stakeholders** – measure agreement between stakeholders and answered concerns, measure understanding of stakeholders based on common language more misunderstanding. For example, there are several stakeholders in the eLearn system: lecturer, students, admin staff, maintainer, manager who all have different concerns. Lecturer and students are concerned whether it is usable, available and fast, while maintainers are more concerned whether the system is modifiable. The system architecture is beneficial when most of the conflicts are resolved and there is mutual understanding among the stakeholders.
- **Earliest design decisions** – Important decisions should be made early to avoid unnecessary cost in later phases. The benefit can be measured by counting the number of early decisions made and how making these decisions reduce the overall cost. It is better to make as much important decisions as possible in the beginning of the project. For example, the architecture of eLearn system will embody the early decisions such as what browser will the system run on, what kind of pre-requisite are needed, and what protocol is used for communication.
- **Defining constraints on an implementation** –architectural design, specification and stated constraints should be reflected and addressed in the implementation phase. We can measure if the constraint is met how useful it was in reducing the cost and enhancing the system quality when meeting that constraint. For example, in eLearn’s architecture, when we enforce developers to use specific IDE such as Eclipse as it supports several good features for debugging and testing.
- **Influencing the organizational structure** – architecture is used as the basis of work-breakdown structure, it measures the units of planning, scheduling, and budget. The benefit can be measured by scope comprehension across multiple parties, the scope of project should

be clear with the described expectations. For example, in Sunway eLearn system, this will be beneficial by having allocation structure which allocates the work breakdown to the development team, which in turn influences the organizational structure of the developers.

- **Enabling evolutionary prototyping** – skeletal system allows problems to be identified early and reduce risk. For example, the benefit can be measured by how using skeletal system can identify potential security issues. For example, Sunway eLearn team can utilize skeletal system to identify if there's security problem at the beginning of the project such as having a guest user accessing critical data.
- **Improving cost and schedule estimates** – early cost and schedule estimates can be done by using the architecture. Its benefit can be measured by the accuracy of the estimates. Top-down knowledge of the system is useful for setting goals and apportioning budget. However, in terms of cost estimates, bottom-up understanding of the system's pieces are normally more accurate than top-down understanding of the system. For example, in Sunway eLearn system, there are two teams where the first team handles features, such as live sessions, and the second team handles the grading system. Every team estimates cost and schedule of their own parts so that the estimates can be accurate.
- **Transferable, reusable model** – the benefit can be measured by the amount of code, requirements, experience and infrastructure that can be reused in a system with similar requirements. When the architectural design can be reused across multiple systems, the development time and cost can be tremendously reduced. For example, online learning platforms have similar architectures in which several architectural designs and decisions are reused, such as authentication mechanisms, live sessions and tests, grading systems. This can speed up the development and allow the team to focus on the unique functionalities of the system.
- **Using independently developed components** – an architecture defines the independently developed components that can be incorporated into the system to reduce cost with higher reliability and flexibility. The benefit can be measured by the amount of the elements that can be incorporated into the system. For example, in Sunway eLearn system, if the live session feature is developed using existing components, the feature can be developed with

reduced time and development cost and higher reliability compared to when developing the feature from scratch.

- **Restricting design vocabulary** – Measure by level of understanding of teams. With less design vocabulary, system designs are easily understood instead of various terms and definitions. For example, the architecture design of eLearn will be used across stakeholders, by restricting the choice of elements and focusing on unified definitions of quality attributes of interest.
- **Basis for training** – Easy to explain the system to new members. Achieved by having different views to explain to different stakeholders that have separate concerns. Benefit is measured by the information represented using different model of views, such as the quality attributes of the system and interaction among attributes. Blackboard Team will be able to utilize models such as component and connector view to explain how each component of the system should interact with one another, for example interactions between runtime elements. Module views can also help the team to understand the structure of the system easily. The learning curve of new members regarding the system should be reduced when employing the architecture for training new members.

- 2- Prioritize the list of thirteen points of benefits of architecture according to some criteria meaningful to you. Justify your answer. Or, if you could choose only two or three of the reasons to promote the use of architecture in a project, which would you choose and why?

Sample Solution: The solution is for open discussion. There is no black/white answer to this question. Every student could discuss this answer differently. What matters most is that this prioritization is supported with a good justification.

- 3- There are several stakeholders of any system. Explain the roles of the following stakeholders and how they are interested in the system architecture (Maintainer, Tester, User, Database Administrator).

Sample Solution:

Maintainer: Responsible for fixing bugs and providing enhancements to the system throughout its life (including adaptation of the system for uses not originally envisioned).

Interest in Architecture: Understanding the ramifications and consequences of a change.

Tester: Responsible for the (independent) test and verification of the system or its elements against the formal requirements and the architecture.

Interest in Architecture: Creating tests based on the behavior and interaction of the software elements.

User: These are the actual end users of the system. There may be distinguished kinds of users, such as administrators, superusers, etc.

Interest in Architecture: Users, in the role of reviewers, might use architecture documentation to check whether desired functionality is being delivered. Users might also use the documentation to understand what the major system elements are, which can aid them in emergency field maintenance.

Database Administrator: Involved in many aspects of the data stores, including database design, data analysis, data modeling and optimization, installation of database software, and monitoring and administration of database security.

Interest in Architecture: Understanding how data is created, used, and updated by other architectural elements, and what properties the data and database must have for the overall system to meet its quality goals.

- 4- Provide an example of a system used by Sunway University. For this system,
 - a. Who are the stake-holders?
 - b. Explain their interest in the system architecture.
 - c. What are the contextual influences on the system architecture?
 - d. What are the business goals that led to the construction of the used system?

Sample Solution:

Example: Sunway University eLearn system

Question 4(a)

The stakeholders of Sunway eLearn system include tester, maintainer, database administrator, architect and end user, such as lecturers and students.

Question 4(b)

The stakeholders' interests:

- **Tester** – Tester tests the system to ensure that it is suitable for use and it is free of bugs/errors. By using system architecture, tester can create tests cases on the behaviour and interaction of the elements in the system. Hence, tester will ensure every behaviour and interaction performs properly. When user enter a wrong password or wrong student ID, eLearn should have an error message and let users enter again. Also, testers are concerned if the required quality attributes and user requirements are achieved. Therefore, the software architecture must allow the implementation to satisfy the requirements of the end users.
- **Maintainer** – Maintainer uses architecture to understand the effect of a change and keep track of all aspect of the system so that they able to maintain the existing code and improve the development. For example, if maintainer change the code without referring architecture, it might cause some data inconsistency.
- **Database Administrator** – Database administrator stores and organizes users' database. Database administrators usually utilize architecture to understand the structure of the data. Database administrators also use architecture to understand how the data is used and created. Therefore, database administrators are concerned about the complexity and security of the data structure. They want to ensure that the data structure is less complex and secured. For example, in Sunway eLearn system, there should be a proper authentication mechanism to ensure the data's security of quizzes and grades.
- **Architect** – Architect concerns about the strategies to achieve all requirements from various stakeholders. Architects needs to create an architecture that can provide a mutual understanding among the stakeholders to satisfy all their needs. For example, in Sunway eLearn system, the end users, which are lecturers and students, want the system to have as many features as possible. Meanwhile, the university management want the system to be developed and launched in a short time with less cost. The architect will then want to think of the way to satisfy all those requirements.

- **End User** – End user requires the system to be functional according to their needs. Users can go through software architecture to understand the provided system functionality and check whether the required functions have been delivered. To end-users, they are interested in the basic terms (not using technical ones) that can help to understand the system functionality; so users can understand the system easily. Users may be concerned with the functional requirements and quality attribute requirements (e.g. reliability, availability, performance). For example, in Sunway eLearn system, the end users (lecturers, students), are concerned about the response time when they join live sessions.

Question 4(c): Contexture influences on the system architecture:

- **Technical context** – technical environment might affect stakeholders' requirement and quality attributes. In today's world, several systems are web-based and cloud-based. Stakeholder have to consider these before listing out the requirements. Basically, Sunway eLearn can only be accessed when there is an internet connection. Indeed, this affects the system architecture, as desktop applications are developed differently compared to web-based applications.
- **Project lifecycle context** – Before designing and developing the system, requirements are gathered and elicited. Maintenance team should perform maintenance referring to the architecture. Testers generates test cases referring to the architecture.
- **Business context** – ensure that the system architecture meet its business goals like save cost and time. Architecture should achieve the organization goals and different stakeholders' goals. For example, the architecture should provide a usable solution to lecturers and students to operate in online academic life. The architecture should allow the university to make a good reputation via providing good online platform.
- **Professional context** – In the process of building a system, the experience and knowledge of architect will directly affect the architect. By applying different standard during development, the system's quality attributes and business goals could be achieved. For example, knowledgeable architect team who knows several languages (e.g. JavaScript, Python), libraries, and coding standards helps to achieve the desired business goals and qualities stated above, and accordingly this affects the quality of architecture.

Question 4(d) Some Business goals:

- **Improve capability/quality of the system** – In Sunway eLearn system, the university will want the system to have high quality and capability. For example, the quality of the system can be improved when the system is easy to use for beginners. Users should be able to understand how to use the system in a short time. For instance, students can see their courses by clicking “courses”. The button is clear and easy to trace. If users try to access eLearn home page without enter correct password and Sunway ID, eLearn will directly bring the user to login page. This enhances the security of the system.
- **Reduce total cost of ownership** – To develop a system with low cost, the system should be developed for ease of installation and repair. In this case, it will reduce the cost of deployment and operation. Since eLearn is an online platform, users can access it through its online link and enter correct student ID and password. eLearn also provide a mobile application and it can be downloaded through Play store (android) and app store (iOS). Not only that, the system should able to transfer its function to a new system when it has retired.
- **Improve market position** – Enhancing reputation is a way to improve market position. Since lecturer have to conduct online class during Covid19 pandemic, the recorded videos for live sessions are easy to be uploaded when lecturers are using blackboard collaborative. Also, students should be able to use microphone and camera in the online class. Providing those features will improve the market position of the system.